

Christophe BOGHOSSIAN



PROFILE

Highly motivated PhD student in Applied Maths at CEA, with a strong background in **Physics**, specializing in **scientific computing, numerical modeling, and applied maths for physics**. Eager to leverage expertise in numerical simulations, numerical schemes, and algorithm evaluation.

CONTACT DETAILS

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🌐 linkedin/Christophe/BOGHOSSIAN
🌐 Christophe BOGHOSSIAN

LANGUAGES

French (Native), **English** (B2/C1),
Spanish (A1/A2)

TECHNICAL SKILLS

◇ Maths, Physics, Numerical modeling, FEM, ML
◇ Python, Fortran90, Matlab, LaTeX, C++

SOFT SKILLS

◇ Scientific rigor, autonomy, and analytical thinking
◇ Comfortable working in multinational environments

ADDITIONAL EXPERIENCE

Private Tutor: Mathematics, Physics, and English basics for high school students.

INTERESTS

Music: Drums, guitar, bass guitar.
Sports: Parkour, Calisthenics, Kung-Fu.

PROFESSIONAL EXPERIENCE

APPLIED RESEARCH INTERN | ONERA, Toulouse 2026.03–2026.08

◇ DEMR Department: Developed and implemented a dispersion error correction method in Fortran90 to optimize Finite-Difference Time-Domain (FDTD) electromagnetic wave propagation simulations.

RESEARCH INTERN | IRAP, Toulouse 2025.03–2025.08

◇ Analysis of large-scale astronomical datasets in Python to estimate physical parameters through deterministic and Monte-Carlo simulation approaches for model validation.

ACADEMIC PROJECTS

TRANSIENT ADVECTION-DIFFUSION & STABILITY ANALYSIS | Aix-Marseille Univ. 2025.01–03

◇ Resolved 1D transient transport equations using Runge-Kutta 4 (RK4) and finite difference discretizations in Python.
◇ Derived Courant-Friedrichs-Lewy (CFL) stability limits, evaluating numerical dissipation, dispersion errors, and spatial state gradients.

ELECTROMAGNETIC FINITE ELEMENT MODELING (FEM) | Aix-Marseille Univ. 2025.01–03

◇ Solved elliptic boundary value problems (BVPs) using Galerkin weak formulations and unstructured meshing in GMSH.
◇ Implemented mixed Dirichlet and Neumann boundary conditions to resolve full spatial electromagnetic field distributions.

CHARGED PARTICLE DYNAMICS | Aix-Marseille Univ. 2024.01–03

◇ Simulated solar particle trajectories in geomagnetic fields by solving Lorentz force equations in Python and Fortran90.
◇ Compared implicit vs. explicit Euler integration schemes to mitigate numerical stiffness and analyzed energy conservation bounds.

FRACTIONAL DIFFUSION & SPECTRAL METHODS | Aix-Marseille Univ. 2023.09–12

◇ Solved 1D fractional-order diffusion equations using finite-difference time discretization in Fortran90.
◇ Evaluated temporal evolution of non-local transport phenomena by transforming system states into Fourier space.

N-BODY TRAJECTORY OPTIMIZATION | Aix-Marseille Univ. 2023.01–05

◇ Formulated and solved non-linear orbital trajectory optimization problems under thruster efficiency and payload constraints.
◇ Implemented multi-variable gradient descent algorithms to compute optimal interplanetary transfer orbits within N-body dynamics.

EDUCATION

MASTER'S IN SPACE TECHNOLOGIES | Toulouse Univ. 2025.09–2026.09

◇ Curriculum focused on space physics, data analysis, and advanced numerical simulation of complex physical systems (constraints on satellites, electronics, ...).

MASTER'S IN FUNDAMENTAL PHYSICS | Aix-Marseille Univ. 2023–2025

◇ Advanced training in theoretical and numerical physics, including statistical mechanics, electromagnetism, and fluid/plasma sciences.
◇ Erasmus Exchange: Bologna, Italy — Specialization in astrophysics.